

Chapter 8

Bank failures and banking crises

Learning objectives

- To define bank and financial failures
- Evaluate strategies to identify problem banks
- Identify successful bank restructuring procedures
- Identify the main causes of banking crises
- Evaluate the costs of banking and financial crises
- Understand the different approaches taken to resolve a banking crisis
- Identify the key regulatory initiatives to reduce systemic risk

8.1 Introduction

Banks are profit-maximising firms and bank management is all about seeking profits and managing risks. Bank managers routinely deal with risk management issues and strategic planning. On occasions, bank managers take on excessive risks or implement value destroying strategies, therefore causing the bank to run into trouble. Like any other firm, banks do fail. The reasons why banks fail are numerous and often interlinked. Managerial deficiencies are often an important reason behind the failure of financial institutions. For this reason, bank risk management is also the focus of regulatory concerns. In addition, because of the potential for systemic risk arising from bank failures, regulators tend to intervene in the banking sector.

Before we discuss the determinants of bank failure and the strategies adopted by regulators to identify problem banks, it is useful to introduce a definition of bank failure. A bank, as a profit-maximising firm, is considered insolvent when its liabilities exceed its assets and its net worth becomes negative. A bank is deemed to have failed if it is liquidated, merged with a healthy bank under government supervision/pressure, or rescued with state financial support. Whereas some think that failing banks should be treated as any other failing firms, the potential for systemic effects arising from bank failures has entailed a varying degree of intervention in the banking system (ranging from deposit insurance to lender of last resort functions of the central bank, to direct state intervention and financing).

How to deal with bank failures is a controversial issue. Banks play such a critical role in the economy that they are subject to more intense regulation than other sectors. As we discussed in Chapter 7, regulation of the banking system is justified by market failures, which can be caused by asymmetric information and negative externalities. Bank regulation can take a number of forms, from deposit insurance to capital requirements (namely, the Basel Capital Adequacy Accords), bank licensing and regular examinations of banks. The main concern of regulators is that the failure of one bank can have a contagious domino effect, leading to the failure of other banks, and this, therefore, can adversely affect the whole financial system. In reality, the transition from the failure of an individual bank to the collapse of a country's banking system is rare (although possible, an example is the collapse of Icelandic banks in 2008). Nonetheless, there have been frequent systemic crises in developed and emerging economies in recent years. These will be analysed in more detail later on in this chapter. We first discuss the determinants of bank failure in Section 8.2. The main strategies used to identify problem banks are explained in Section 8.3, with a focus on early warning systems for bank soundness and the recently introduced stress tests. Section 8.4 presents the key issues of bank restructuring and will discuss the regulatory toolkit. Section 8.5 discusses the causes and consequences of banking and financial crises and Section 8.6 concludes the chapter.

8.2 The determinants of bank failure

It is rare to find a single reason for a bank's failure; rather there are often a number of contributing factors. Nevertheless, a clear understanding of the main reasons as to why a financial institution has run into trouble is necessary to enable regulators to use appropriate tools and ensure success of the proposed solutions. To give a very simple example, if a bank is facing a temporary liquidity problem, the extension of a liquidity line by the central bank (or access to the discount window) might help in solving the bank's problem. However, if the bank faces solvency rather than liquidity problems, then allowing the bank to access the discount window is unlikely to solve the bank's problems and it can have costly consequences for the taxpayers.

The academic and policy literature identifies the key determinants of bank failure as follows:

- poor management;
- fraud;
- regulatory forbearance;
- too big to fail;
- clustering;
- macroeconomic and systemic factors.

8.2.1 Poor management

Deficiencies in the management of banks are a contributing factor in virtually all cases of bank troubles. In most bank failure cases, the bank senior management will appear as the culprit. For example, in the case of Barings in 1995, even though the bank was eventually brought down by a 'rogue trader', the underlying problem was bad senior management.

Indeed, head office allowed Nick Leeson to run both the front and the back office of the Singapore branch simultaneously, despite an internal audit report recommending that the trader should stop managing the back office. This allowed Mr Leeson to hide huge losses. More recently, the trading incident at Société Générale in January 2008, which involved the trader Jerome Kerviel and resulted in a loss of €4.9 billion – the largest in banking history, also calls Société Générale's management into question (see Box 8.2).

A 2011 report by the UK Financial Services Authority (FSA), which aimed to identify the multiple factors resulting in the failure of the Royal Bank of Scotland (RBS) in October 2008 identifies:

errors of judgement and execution made by RBS executive and management, which in combination, resulted in RBS being one of the banks that failed amid the general crisis. These were decisions for whose commercial consequences RBS executive and Board were ultimately responsible.

(Financial Services Authority, 2011a)

The FSA report concludes that, even taking into account deficiencies in regulation and the macroeconomic conditions at the time, RBS's failure ultimately resulted from poor decisions made by the management and Board of Directors (see also Box 13.2 on the failure of RBS).

Beside these extreme examples, poor management can cause asset-side problems (e.g. bad loans, investment losses), liability-side problems (e.g. liquidity problems, deposit withdrawals, bank runs) and off-balance-sheet problems (e.g. derivative losses). Poor asset management, or a weak loan portfolio because of excessive exposure in one or more sectors, can eventually lead to failure.

The IMF identifies five types of loan performance categories for external reporting purposes:

- 1 *Standard*. Credit is sound and payments current.
- 2 *Watch*. Subject to conditions that if uncorrected, could raise concerns about full repayment.
- 3 *Substandard*. Full repayment is in doubt due to inadequate protection. Interest or principal overdue (90 days +).
- 4 *Doubtful*. Assets for which collection is considered improbable. Interest or principal overdue (180 days +).
- 5 *Loss*. Virtually uncollectible. Interest or principal overdue (1 year +).

Substandard, doubtful and loss are considered as **non-performing loans (NPLs)** and remain so until either the loan is written off or until principal and interest payments are received. More specifically, the IMF (2004) defines non-performing loans as loans on which debtors have failed to make contractual payments for a predetermined time.¹

It should be noted that a loan classified as non-performing does not necessarily lead to losses. If there is adequate collateral, losses might not occur. Conversely, loans may be lost even though they were never classified as non-performing.

¹ The International Monetary Fund posted on its website a 'Guide' ('Compilation guide on financial soundness indicators') in 2004 (and then published it in 2006 and revised in 2007 and in subsequent years) with the purpose of providing information and advice on concepts and definitions, as well as sources and techniques, for the compilation and dissemination of financial soundness indicators. See IMF (2007, 2019).

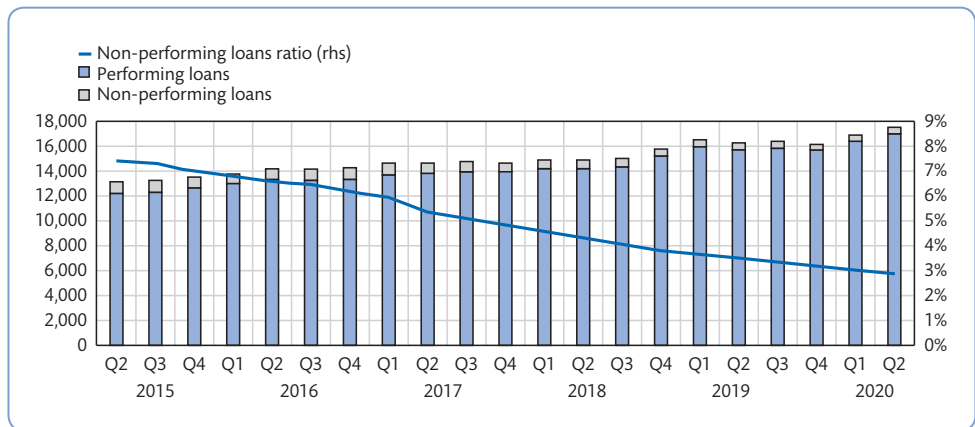


Figure 8.1 Eurozone banks' NPLs 2015Q2–2020Q2 (€bn, %)

Source: ECB (2020) Supervisory Banking Statistics. Available at <https://www.bankingsupervision.europa.eu/banking/statistics/html/index.en.html>

Not all countries adopt the same definition of NPLs, and there may even be different definitions in use within a single country depending on the sector involved (financial institutions, quoted corporations, small enterprises, government entities and so forth). In addition, international accounting and banking standards refer to loans being *impaired* rather than non-performing.² Proper recognition and provisioning for NPL are essential for crises management and prevention. However, while there is broad consensus on the need for more rigorous loan classification rules, there is some controversy over the timing and tightening of rules. The fear is that markets might overreact to the full disclosure of NPLs, particularly in emerging markets or in markets where the existing economic conditions are difficult. Yet suspicions that the true scale of the problem is being hidden can hurt market confidence. Figure 8.1 illustrates the trend in NPLs in the EU following the sovereign debt crisis.

From Figure 8.1, we can see that the credit quality of loan portfolios improved: the aggregate NPL declined to stand at 2.94 per cent, the lowest level since in 2015. This improvement in NPLs was remarkable as the NPL to total assets ratios of EU banks had remained relatively stable at around 4 per cent of total loans since the global financial crisis. It is worth mentioning, though, that the average EU-level statistics mask some substantial cross-country differences, with NPL ratios varying between 0.70 per cent in Luxembourg and 30.31 per cent in Greece (Q2, 2020). These encouraging results are likely to be reversed by the impact of the Covid-19 pandemic, as regulators are implementing various measures to help the banking sector (see Box 8.1).

An essential step in any bank restructuring programme is to measure correctly the amount of non-performing loans. This is a major task, due to the varying practices of loan classification and different regulatory environments. Some regulatory authorities use quantitative criteria, such as the number of days loan repayments are overdue; others rely on qualitative norms such as the clients' financial status, or on management judgement about future loan repayments.

² Impairment is a specific term used in the International Accounting Standard 39 (IAS 39) and by the Basel Committee on Banking Supervision (BCBS).

BOX 8.1 EU BANKS URGED TO PREPARE FOR BAD LOANS AS PANDEMIC HITS ECONOMY

Head of body responsible for winding down failing lenders issues warning

European banks need to prepare their balance sheets for the risk of pandemic-induced non-performing loans hitting them in the new year, the head of the EU agency tasked with winding down failing lenders has said.

Elke König, chair of the Single Resolution Board, rejected suggestions from the European Central Bank that the EU needs to set up a network of “bad banks” to handle higher non-performing loans (NPLs), but she warned banks needed to do intensive work to sort out viable loans from unviable ones.

In an interview with the *Financial Times* she also urged the EU to properly harmonise its state aid rules for handling embattled banks with its “resolution” regime, which intervenes to ward off systemic financial crises. Ms König argued that as things stand there was a “misalignment” in the system.

The SRB was set up following the eurozone sovereign debt crisis to wind down stricken banks while securing financial stability and consistent treatment of the region’s lenders. But with regulators now weighing the risks of a surge in NPLs, the agency is still working with an incomplete system of EU bank-crisis rules which must operate over a patchwork of different national arrangements.

Last month, the ECB’s top bank supervisor Andrea Enria wrote in the FT that, in a “severe but plausible” scenario, non-performing loans at eurozone banks could reach €1.4tn, well above the levels of the 2008 financial crisis and the ensuing EU sovereign debt crisis.

Ms König said it was too soon to know how bad the situation with NPLs would get, because this would depend on the nature of the downturn and recovery. Current actions by governments, such as state guarantee schemes, were “shielding” the lenders, she added.

But she said that non-performing loans could start coming through in the first and second quarters of next year. In light of this, her message to banks was “be aware NPLs are coming and the

best thing to do is address them early ... That is the best thing we can do for the time being, and then it is steering through the fog.”

The European Commission will next month set out its own proposals for handling what Valdis Dombrovskis, executive vice-president overseeing economic policy, has said will be a “likely rise” in NPLs. These include further developing secondary markets in which NPLs are bought and sold, and reforms to insolvency and debt-recovery frameworks.

The ECB’s Mr Enria has separately advocated the creation of either a European “asset management company”, or a network of them, to take on NPLs. Ms König said that she was cautioning against this idea, and that banks doing their “homework” was the easier way forward.

“The entire debate ... always lacks one component: who is footing the bill,” she said. Ms König added that “it feels sometimes as if this is the magic system,” one where losses are supposed to evaporate, something “that is not going to happen”.

Banks need to “get the full transparency” of these portfolios, she said. “They need to be able to separate a portfolio and to set out how they would like to service it, what would be their idea, how would they deal with it.”

She added that in the current crisis there was still a market for bad real estate and corporate loans. When it comes to loans to small and medium-sized enterprises and retail customers, these are traditionally “professionally managed by a bank as an ongoing relationship”.

Ms König underlined the extensive work that has been done since the dark days of the eurozone crisis to strengthen the resilience of banks, and to ensure that they can be safely wound down if they fail. This work has included making sure money can be raised by wiping out bank bondholders, so shielding taxpayers from footing the bill.

She noted that one of the tools available to her agency was an “asset separation” power to strip bad loans out of broken banks’ balance sheets.



Box 8.1 EU banks urged to prepare for bad loans as pandemic hits economy (continued)

But she emphasised that the EU system for handling failing banks was still a work in progress. One step she has called for is the creation of a “pre-liquidation tool” allowing her agency to intervene earlier in a crisis to save the good part of a bank, preventing needless destruction of value.

The SRB and the broader system it oversees has been fully up and running since 2016, and in that time has handled one bank failure: Spain’s Banco Popular.

One ongoing problem, she said, was that the conditions attached to banks put through national insolvency proceedings differed from those

attached to interventions by her agency. This issue flared up in 2017 when two regional Italian banks received state aid as part of bankruptcy proceedings while their senior creditors were shielded from losses.

This prompted complaints at the time from Berlin and some other capitals that the measures amounted to a loophole allowing nervous governments to get around the principle adopted by the EU in the wake of the financial crisis that investors, including senior creditors, should face losses before the taxpayer.

The European Commission is set to review the bank resolution system next year.



Source: Fleming, S. and Brunsden, J. (2020) ‘EU banks urged to prepare for bad loans as pandemic hits economy’, *Financial Times*, 11 November.

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An area of concern during bank restructuring is the valuation of **collateral**. In theory, most bank loans are collateralised (typically on real estate) and this should provide available resources to any restructuring agency. In practice, collateral is often worth considerably less than book value and can only be recovered if bankruptcy procedures operate efficiently. As aggregate demand weakens during banking crises, collateral values (such as property prices) drop steeply. Moreover, a large number of simultaneous ‘fire sales’ may force the value of collateral to drop even further. This raises the question of how long a restructuring agency should hold the assets of distressed banks. Finally, the value of the collateral depends also on the credibility of the legal process to enforce repayments.

8.2.2 Fraud

Fraud has long been recognised as a key cause of bank failure. A 1986 study commissioned by the American Bankers Association to identify the main reasons for US bank failure noted (Benston *et al.*, 1996): ‘Because fraud has been the single most important cause of bank failures, both in the past and in the present, it is distinguished from excessive risk-taking.’

Along similar lines, Barker and Holdsworth (1993) reported a study by the US House Committee on Government Operations that found about 50 per cent of bank failures and 25 per cent of thrift failures in the 1980s were principally due to fraud. However, it is often difficult to secure a conviction because of the fine line between fraud and bad management. Box 8.2 revises some of the largest episodes of fraud in financial institutions.

8.2.3 Regulatory forbearance

When financial intermediaries run into trouble, regulators may be under pressure to not apply existing regulations too strictly. As discussed in Section 7.5, this is known as regulatory forbearance and it arises when applying regulations (for example, imposing compliance with

BOX 8.2 ROGUE TRADERS AND BANK LOSSES

In 1995 Mr Nick Leeson was the rogue trader who brought down Barings Brothers, a British merchant bank, by uncovered exposures in the derivatives market. He went into the red by \$1.2bn by trading on Asian markets, after what he claimed was a well-intentioned attempt to cover up losses in a client's account. He was jailed for fraud.

Also in 1995, Mr Toshihide Iguchi at Daiwa Bank in Japan lost around \$1.1bn (after more than 10 years of illicit trading) while dealing in US Treasury bonds. Daiwa Bank did not fail because it sold its assets and had considerable reserves; however, it was a massive hit for its reputation. Mr Iguchi was fined \$2.6m and sentenced to four years in prison.

In 2002 Mr John Rusnak caused losses of some \$750m to Allied Irish Bank, Ireland's largest bank, in unauthorised foreign exchange dealing at its American subsidiary, Allfirst. Mr Rusnak expected the yen to strengthen against the dollar but this did not occur. The bank thought that options contracts were purchased as the internal system showed. Instead they weren't actually bought. Therefore, there was no insurance against the loss. The bank's solvency was not threatened but the bank absorbed the losses at a price of a significant reduction in earnings and capital.

Another rogue trader, Mr Peter Young, a fund manager of investment bank Morgan Grenfell Asset Management, a Deutsche Bank company, lost some \$380m from the funds he ran, after hiding a series of unauthorised investments. Deutsche Bank had to inject \$300m in cash to replace the fund; however they incurred huge losses after a third of the investors left the fund within a few weeks.

In 2002 Central Europe had its own rogue trader in Mr Eduard Nodilo, a dealer at Rijecka Banka, Croatia's third-biggest bank, which accumulated \$98m in foreign exchange losses wiping out the bank's capital.

In January 2004, Australia's biggest bank, the National Australian Bank, revealed that some of its foreign currency options traders in Melbourne and London had engaged in unauthorised trading. Losses have been estimated at \$445m.

In January 2008, the trading incident at Société Générale, involving the trader Jerome Kerviel, resulted in a loss of €4.9bn. This event sparked the question as to whether it was a case of an isolated rogue trader or a more complex case of a 'rogue business model':

Société Générale uncovered a fraud, exceptional in its size and nature: one trader had taken massive fraudulent directional positions in 2007 and 2008, beyond its limited authority. Aided by its knowledge of control procedures, he managed to conceal these positions through a scheme of elaborate fictitious transactions [...] which eventually resulted in the largest trading loss in banking history.
(*Financial Times*, 24 January 2008)

A similar question was raised in 2011 in the case of Kweku Adoboli's actions at UBS:

UBS has discovered a loss due to unauthorized trading by a trader in its Investment Bank. The matter is still being investigated, but UBS's current estimate of the loss on the trades is in the range of USD 2 billion. It is possible that this could lead UBS to report a loss for the third quarter of 2011. No client positions were affected.
(*Financial Times*, 15 September 2011)

In November 2012, Adoboli was jailed for the UK's biggest bank fraud, while UBS was fined £29.7 million over its conduct in the case.

The worlds of securities and derivatives trading have seen increased risks brought about by high-frequency trading and automated trading strategies, and institutions are bracing themselves for the next scandal. While individual 'rogue traders' have not hit the headlines recently, between 2008 and 2018 a larger than ever number of scandals and fraud episodes led to the world's largest banks being hit with an unprecedented number of misconduct fines, amounting to around £380 billion (Centre for Banking Research, 2020).

capital adequacy ratios) could worsen the institution's problems and could result in failure, possibly triggering a run on other financial institutions. Regulatory forbearance itself may even cause a banking crisis as bank regulators may become locked into an ever-worsening spiral, resulting in a loss of public confidence in how banks and the financial system in general are being regulated. Regulatory forbearance for poorly managed banks may result in damage to the regulator's credibility and authority. In addition, the moral hazard risk associated with regulatory forbearance could provide incentives for excessive risk taking.

In the context of bank failures, when bank auditors and regulators 'miss' important signals from the distressed bank, or they put the interests of the regulated bank ahead of taxpayers, they are 'guilty' of forbearance. In many cases of failures, subsequent investigations show that exposure limits were exceeded with the knowledge of the regulators – suggestive of widespread forbearance in dealing with troubled banks.

8.2.4 Too big to fail (TBTF)

'*Too big to fail*' refers to the viewpoint whereby the largest banks in a banking system are viewed as being too big to be allowed to fail and therefore they have guaranteed access to the lender of last resort. The idea of the TBTF is not new – the term became used in the 1980s following the bailout of the US bank Continental Illinois. Since the 1980s though, financial institutions have grown much larger and more interconnected. Indeed it is often argued that size by itself is not the relevant criteria for bailing out troubled banks – rather it is the significance or importance of banks in specific markets and the expected scale/impact of potential failure. This is what prompted the rescue of US investment bank Bear Stearns (in March 2008) and the government bailout of American International Group (AIG) (in September 2008), once the world's largest insurer. This has prompted the use of other acronyms, such as too systemic to fail (TSTF) or too interconnected to fail (TITF).

While no financial regulatory authority will ever provide guaranteed access to lender of last resort financing, history does tell us that (for systemic and other reasons) large banks are likely to be bailed out more than small banks. The policy of the TBTF applies, to a certain degree, in all countries. Critics argue that the implicit government guarantees for the biggest banks prompt them to take excessive risks and therefore make them more likely to fail.

Following the 2007–2009 global financial crisis, the concept of TBTF has emerged from the regulatory and academic debate and has moved into the public arena and there have been calls for breaking up big banks. For example, the 2010 Dodd–Frank Wall Street Reform and Consumer Protection Act (Dodd–Frank) is the US government's attempt to resolve the TBTF problem for US banks. At the global level, in 2011 the Financial Stability Board identified an initial group of 29 banks as systemically important financial institutions (SIFIs), to be subjected to more stringent prudential standards. The G-SIBs group is revised annually and included 30 institutions in 2020 (see also Chapter 7). Similarly, the BCBS set forth an additional capital requirement for global systemically important banks (G-SIBs).

8.2.5 Clustering

Clustering refers to the fact that bank failures in a country tend to be clustered around a few years rather than being spread evenly over time. The presence of a so-called 'herd instinct' or 'contagion effect' among depositors and investors may help explain a run on several banks

over a short period of time. However, macroeconomic factors are also important in explaining clustering effects.

Another reason for clustering may relate to the failure of timely intervention by the government or regulatory authorities. The Japanese banking crisis of the 1990s offers an example of the interaction of macroeconomic factors and badly timed regulatory intervention.

8.2.6 Macroeconomic and systemic factors

There are a number of factors that do not slot easily in any of the previous categories. There is a vast literature on the causes of bank failures and banking crises; this literature is fairly divided between advocates of microeconomic factors and supporters of macroeconomic reasons and systemic factors. These are discussed in more detail in Section 8.5.3.

Qualitative reviews of bank failure, as the ones described in this section, provide some insight into what causes a bank to fail. Quantitative models provide econometric models of bank failure, borrowing from the literature on corporate bankruptcy. These models aim to measure bank soundness and provide early warning indicators of trouble. These early warning systems (EWS) are discussed in Section 8.3.

8.3 Early warning systems for bank soundness

Recent episodes of turmoil in the international financial markets have called attention to the need for better tools to monitor bank risk taking as well as the vulnerabilities in the financial system. From a regulator's perspective, the key is to anticipate the next bank failure by identifying those institutions that display underlying vulnerabilities, taking into account potential triggers in the wider economic environment (for example, contagion from other banks/countries, trade shocks, political instability, etc.).

Put simply, **early warning systems (ESW)** are models designed to draw regulators' attention to certain key variables associated with past crisis. These variables can reflect the risk of a single financial institution (micro-prudential approach) or the risk of the financial system as a whole (macro-prudential approach).

Bank supervisors have traditionally concentrated on assessing the risk profile and soundness of individual financial institutions. This micro-prudential approach has focused mainly on:

- 1 Standard balance sheet and income statement financial ratios. This includes the so-called CAMEL variables.
- 2 Market prices of financial instruments, such as bank stocks and subordinated debt.
- 3 Measures of bank risk and financial strength, such as deposit rates.

In addition, regulators also collect key macroeconomic indicators. However, critics say that the emphasis on micro surveillance prior to 2007 failed to spot the risk arising from the growing interconnectedness of financial institutions, both domestically and across borders.

Although the use of EWS is not new in policymaking, the 2007–2009 financial crises prompted national regulatory authorities and international organisations, such as the IMF and the FSB, to work together to revise the existing tools to anticipate systemic problems.

Among these initiatives are the ongoing efforts to develop and use macro-prudential indicators, defined broadly as indicators of the health and stability of financial systems.

EWS can be generally divided in models based on financial ratio analysis or peer group analysis and statistical models (for example, models predicting failure or survival rates, models estimating ratings and the probability of rating downgrades, and models estimating expected losses).

8.3.1 Financial ratio analysis based models

Financial ratio analysis is an important tool for bank management and it is used both internally and externally to evaluate bank performance – it investigates areas such as profitability, asset quality, liquidity, solvency and capital adequacy (see Section 9.4 for details on bank financial ratio analysis).

In terms of EWS, a bank financial condition is related to a set of key financial ratios, which should be within a certain range for the bank to be operating in a safe and sound manner. If one or more of these key financial ratios exceeds a predetermined critical level or lies within a set interval, then the EWS generates a warning signal, a ‘red flag’ that indicates that more attention should be paid to that particular bank.

In this context, a bank's performance is benchmarked both against its past performance (i.e. checking whether a particular ratio – or set of ratios – is an outlier with respect to the bank past performance) and/or against the performance of a *peer group* of banks. A peer group is normally defined based on some common characteristics, such as bank size (large banks vs. small banks), bank specialisation (co-operative banks, saving banks, commercial banks, investment banks) or other characteristics that might be relevant in some countries (domestic vs. foreign banks; listed banks vs. non-listed banks; state owned vs. privately owned banks).

Peer group analysis is undertaken on the basis of financial ratios for a group of banks together. This is then used to establish whether an individual bank is performing in a significantly different manner from its peers, with a view to identifying the reasons for such significant difference, which may or may not imply supervisory concerns. Within each peer group, a first step of peer analysis is a simple identification of the best/worst performers compared with the peer average. In addition, financial ratios are ranked from best to worst, and percentile rankings are calculated with the aim of providing regulators an overall view of a bank performance compared to similar institutions. Individual banks whose financial ratios have deteriorated relative to the averages of their respective peer group can then be identified and action taken if necessary. Some regulators follow a formalised procedure for the use of financial ratio analysis for regulatory purposes. The best known of these models is the CAMELS rating, adopted by the US Federal Reserve.

8.3.2 CAMELS rating

In the US, bank supervisors rate an individual bank's overall safety and soundness to produce what are commonly referred to as **CAMELS** ratings. CAMELS is the abbreviation for the components of a bank's condition that are assessed: **C**apital adequacy; **A**set quality; **M**anagement quality; **E**arnings; **L**iquidity; **S**ensitivity to market risk.

Table 8.1 summarises the key financial ratios used to evaluate a bank's financial condition. The Fed uses a mix of publicly available information (from bank financial statements) and

Table 8.1 CAMELS rating

CAMELS	Variables
Capital adequacy	Total risk-based capital ratio; Tier 1 ratio; charge offs to loan loss reserves
Asset quality	NPLs/total assets; non-current loan ratio; loans secured by commercial real estate/total assets; other loans/total assets
Management quality	Non-interest expenses to revenue (net interest income plus non-interest income)
Earnings	Return on assets (ROA), net interest margin (NIM)
Liquidity	Core deposits/total assets; volatile liabilities ratio (long-term assets/short-term liabilities)
Sensitivity to market risk	Non-interest income/total assets (proxy returns on risky assets)

private information supplied by bank management to assign a composite rating. CAMELS ratings range from 1 (best rating) to 5 (worse rating). These ratings are disclosed only to bank senior management and are not made public, except in the event of a bank's failure.

Figure 8.2 illustrates the average and distribution of CAMELS ratings of US banks over the past 20 years. Strong banks have a CAMELS rating of 1 or 2, while weak banks have ratings of 3, 4 and 5.³ The data show that bank average ratings were low during the credit crunch of the early 1990s, then recovered during the late 1990s and early 2000s, only to deteriorate quickly from 2007 onwards.

Figure 8.3 shows how the pattern in the average rating reflects the changes in the number of banks rated in each of the five CAMELS rating categories. More specifically, the data show a larger decrease in 'strong banks' (particularly banks achieving the highest rating of 1) mirrored by an increase in banks rated worse with a rating of 5. Also worrying for regulators is the steady increase in the percentage of weak banks with a rating of 3, which may indicate a worsening of financial condition for banks across the system.

Since the global financial crisis, US regulators have also introduced **stress testing** as a supervisory tool. The first stress testing exercise, known as the Supervisory Capital Assessment Programme (SCAP) was conducted in 2009. After the SCAP, US regulators introduced two distinct but related supervisory programmes that rely on stress testing: (i) DFAST – the so-called Dodd–Frank Act stress tests, which aims to quantitatively assess how bank capital levels would fare in stressful economic and financial scenarios; and (ii) CCAR (Comprehensive Capital Analysis and Review), which combines the quantitative results from the stress tests with more qualitative assessments of the capital planning processes used by banks.

The use of stress testing has become a popular regulatory tool in many countries, as the global financial crises shifted the focus of regulation from the risks faced by individual financial institutions to systemic risk.

³ More specifically, a composite CAMELS rating of 1 means that the bank is basically sound in every respect. A CAMELS rating of 2 means that the bank is fundamentally sound, but may have modest weaknesses correctable in the normal course of business. A CAMELS rating of 3 means that a bank has a combination of financial, operational, or compliance weaknesses ranging from moderately severe to unsatisfactory. A CAMELS rating of 4 means that a bank has a moderate volume of serious financial weaknesses or a combination of other conditions that are unsatisfactory to supervisors. Finally, a CAMELS rating of 5 means that the bank has an extremely high immediate or near-term probability of failure (Bassett *et al.*, 2012).

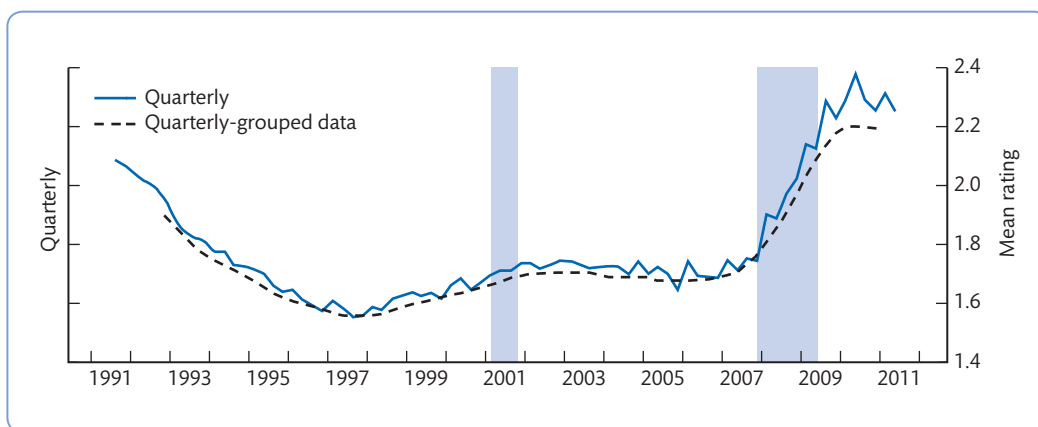


Figure 8.2 Average and distribution of CAMELS rating, 1990–2011

Source: Bassett *et al.* (2012), p. 8.

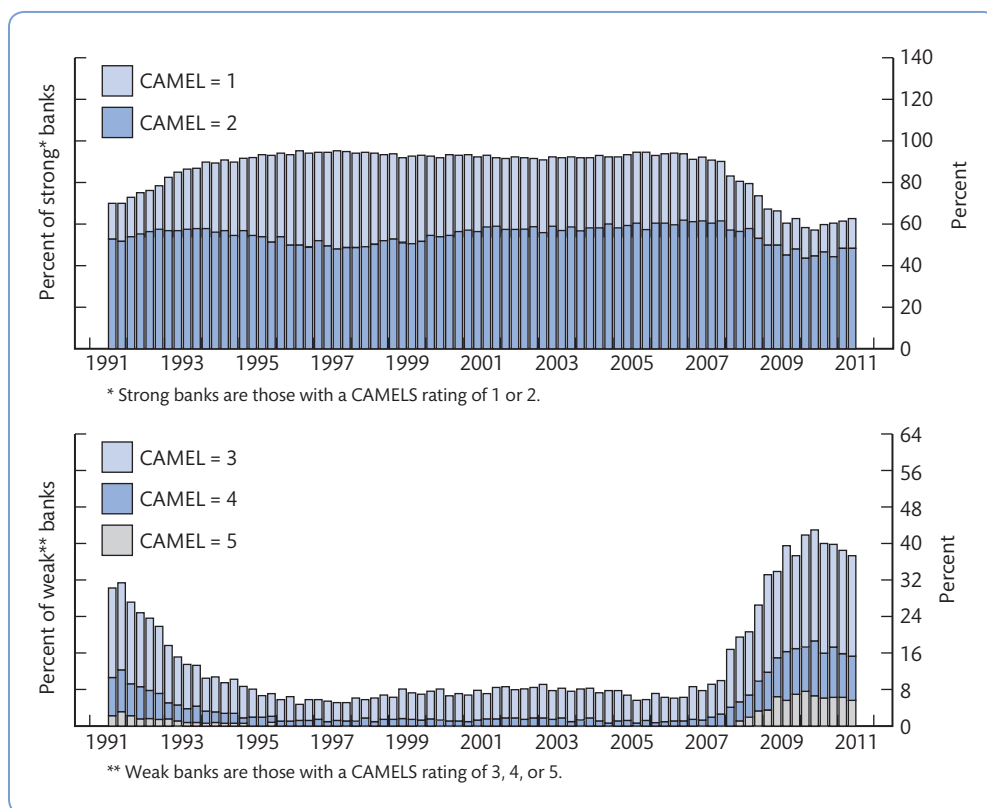


Figure 8.3 Percentage of strong and weak banks, 1990–2011

Source: Bassett *et al.* (2012), p. 8.

8.3.3 Stress testing

In recent years, stress testing has grown in importance, due to increased regulatory requirements and an uncertain economic environment in many countries.

In general terms, stress testing is designed to complement standard Basel capital ratios by adding a more forward-looking perspective and by helping to ensure that banks will have enough capital to keep lending even under highly adverse circumstances. In addition, the disclosures of stress test results can help promote market discipline by providing consistent and comparable information about banks' financial conditions. The Basel Committee on Banking Supervision introduced stronger testing guidelines in 2009 with the objective of ensuring that financial institutions are able to meet capital and liquidity requirements under stressed conditions. These guidelines have been followed by those from the CEBS (Committee of European Banking Supervisors) now European Banking Authority (EBA) in 2010, as discussed in Box 8.3.

BOX 8.3 THE EUROPEAN BANKING AUTHORITY AND EU-WIDE STRESS TESTS

The European Banking Authority (EBA) is an independent EU Authority that works to ensure effective and consistent prudential regulation and supervision across the European banking sector. Its overall objectives are to maintain financial stability in the EU and to safeguard the integrity, efficiency and orderly functioning of the banking sector (see www.eba.europa.eu/).

The EBA was established in November 2010 (Regulation Number 1093/2010 of the European Parliament and of the Council of 24 November 2010) and officially came into being on 1 January 2011. It took over all existing and ongoing tasks and responsibilities from the Committee of European Banking Supervisors (CEBS).

The EBA has some broad competences, including strengthening international supervisory co-ordination, promoting supervisory convergence and providing advice to the EU institutions in the areas of banking, payments and e-money regulation. Its main task is to contribute to the creation of the *European Single Rulebook* in banking, which provides a single set of harmonised prudential rules for financial institutions throughout the EU.

The EBA is part of the *European System of Financial Supervision* (ESFS), which comprises three supervisory authorities: the European Securities and Markets Authorities (ESMA), the European Banking Authority (EBA) and the European Insurance and Occupational Pensions Authority (EIOPA). The ESFS also comprises the European Systemic Risk Board (ESRB) and the Joint Committee of the European Supervisory Authorities as well as the national supervisory authorities of EU member states.

The main decision-making body of the EBA is the Board of Supervisors. It is currently chaired by Mr Jose-Manuel Campa; EU member states are represented by officials of the national central bank. The Board of Supervisors also includes observers from the European Commission (EC), the European Systemic Risk Board (ESRB), the European Central Bank (ECB) and other regulatory authorities that are part of the ESFS. The EBA is headquartered in the City of London.

The EBA main competencies are in three broad areas: (i) regulation; (ii) oversight; (iii) consumer protection.

Within its main competencies, one of the responsibilities of the EBA is to ensure the stability of the financial system in the EU. To this end, the EBA is mandated to monitor and assess



Box 8.3 The European Banking Authority (*continued*)

market developments as well as to identify trends, potential risks and vulnerabilities stemming from the micro-prudential level. One of the EBA's primary supervisory tools is the EU-wide stress test exercise.

In co-operation with the European Systemic Risk Board (ESRB), the EBA conducts EU-wide stress tests in a bottom-up fashion, using methodologies, scenarios and key assumptions developed by the EBA in co-operation with the ESRB, the European Central Bank (ECB) and the EU Commission.

The aim of such tests is to assess the resilience of financial institutions to adverse market developments, as well as to contribute to the overall assessment of systemic risk in the EU financial system. In the first EU-wide stress test exercise in 2011 they tested 90 banks in 21 countries, covering 65 per cent of banking assets and at least 50 per cent of banking assets in each country. The results of the exercise were made public and provided an unprecedented level of transparency on banks' exposures and capital composition.

The results of stress tests exercises are made publicly available on the EBA website.

EBA announces the 2021 EU stress-test exercise

Following the postponement of the 2020 exercise, due to the Covid-19 pandemic, the 2021 EU-wide stress test will provide valuable input for assessing the resilience of the European banking sector. The EU-wide stress test will be conducted on a sample of 50 EU banks – 38 from countries under the jurisdiction of the Single Supervisory Mechanism (SSM) – covering roughly 70 per cent of total banking sector assets in the EU and Norway, as expressed in terms of total consolidated assets as of end 2019. Given the specific macroeconomic conditions caused by the Covid-19 pandemic coupled with a high degree of uncertainty this year, the focus on the different objectives will depend on the conditions closer to the publication date.

Accordingly, the adverse scenario is based on a narrative of a prolonged Covid-19 scenario in a 'lower for longer' interest rate environment, in which negative confidence shocks would prolong the economic contraction. The EBA expects to publish the results of the exercise by 31 July 2021 (www.eba.europa.eu/risk-analysis-and-data/eu-wide-stress-testing).

Despite their growing popularity, stress tests also raise a number of issues, including problems with data collection, the use of different methodologies and need to reconcile internal expert judgement and external benchmarks as well as national and international regulatory requirements. From a bank's point of view, stress testing can be a very expensive and time-consuming exercise; banks criticise both the methodologies and data used. The long-term aim of regulators is to embed stress testing in bank risk management practices rather than it simply being a regulatory tool.

8.3.3.1 Different approaches to stress testing

There are different approaches to stress testing, the main ones being:

- 1 the bottom-up approach (BU);
- 2 the top-down approach (TD);
- 3 the reverse approach.

In a *bottom-up* approach, banks conduct regular evaluations of their positions relative to a set of common scenarios (provided by the authorities). This is the approach followed by the EBA. One of the advantages of the BU tests is their use of extremely granular information

on individual banks' trading portfolios and overall exposures. This permits a more detailed insight into how an individual bank might be affected by worsening macroeconomic and market conditions. On the flip side of the coin, BU tests are institution-specific and therefore it is difficult to compare results across banks.

In the *top-down* approach, the regulatory authorities set the macroeconomic scenarios and conditions under which the test should be run, and calculate the results without the involvement of the banks themselves. The focus with top-down tests is as much on the banking system as a whole as it is on individual institutions. A TD stress test first draws on macroeconomic scenarios and on statistical estimates of the impact of adverse economic conditions on credit and market exposures. Second, it builds formal 'maps' of transmission of economic shocks onto the financial system. By applying the same scenario, on the same model, with the same assumptions and at the same time, TD tests allow for comparisons across banks, as well as offering a framework for understanding and identifying particular areas of vulnerability in the banking system as a whole. In addition, TD models lack the balance sheet granularity of the BU models. The TD approach is favoured by many central banks, including the Bank of England, whose RAMSI model (Risk Assessment Model of Systemic Institutions) is an example of a top-down stress testing model and is one part of the Bank's risk assessment toolkit.⁴

Reverse stress tests require a bank to assess scenarios and circumstances that would render its business model unviable, thereby identifying potential business vulnerabilities. Reverse stress testing starts from an outcome of business failure and identifies circumstances where this might occur. This is different to general stress and scenario testing which tests for outcomes arising from changes in circumstances. Reverse stress tests were introduced by the UK Financial Services Authority in December 2012 and are now the remit of the Prudential Regulatory Authority (PRA) within the Bank of England. Reverse stress tests are to be used by banks as a risk management tool to help managers to overcome 'disaster myopia' and improve contingency planning.

8.4 Bank restructuring

Despite regulators' efforts to identify vulnerabilities in the financial systems, bank failures do happen. A crucial question on regulators' mind is 'What is the best policy option for rescuing a troubled bank?' In this section we are going to discuss the tools in the regulatory toolkit to deal with bank failure.

Bank restructuring has many (sometimes conflicting) aims and as a consequence, there is no universally accepted way to carry out a successful restructuring exercise. The ultimate goal of bank restructuring is assumed to be a lower probability of the bank's default with a minimal taxpayer burden (Landier and Ueda, 2009).

Before we start discussing the bank resolution mechanisms available to policy makers, we need to introduce some basic bank resolution terminology. Some terms related to bank resolution have a range of meanings and different terms are used in different countries. The terminology presented in Table 8.2 is used in most of the IMF literature on banking crises and resolution.

⁴ For more details on the Bank of England RAMSI model see Burrows *et al.* (2012).

Table 8.2 Bank resolution terminology

Intervened bank	An insolvent or non-viable bank where the authorities have taken over the powers of management and shareholders. Such a bank may be closed or may stay open under the control of the authorities while its financial condition is better defined and decisions are made on an appropriate resolution strategy.
Resolved bank	A bank undergoing a resolution strategy, including liquidation, merger or sale, transfer to a bridge bank, recapitalisation by the government, and sales or transfers of blocks of assets or liabilities.
Bank closure	When a bank ceases to carry on its business as a legal entity. Withdrawal of the banking license typically accompanies a closure. A closure may be part of a legal process of achieving the orderly exit of a weak bank through a range of resolution options, including liquidation or a complete or partial transfer of its assets and liabilities to other institutions.
Bank liquidation	The legal process whereby the assets of an institution are sold and its liabilities are settled to the extent possible. It can be voluntary or forced, within or outside general bankruptcy procedures, and with or without court involvement. In liquidation, assets are sold to pay off the creditors in the order prescribed by the law. Liquidation can take place under a country's bankruptcy or company laws or pursuant to a <i>special resolution regime (SRR)</i> for banks.
Merger (or sale)	When all the assets and liabilities of the firm are transferred to and absorbed into another institution. Mergers can be voluntary or government assisted.
Purchase and assumption (P&A)	When a solvent bank purchases all or a portion of the assets of a failing bank, including its customer base and goodwill, together with all or part of its liabilities. In a P&A operation, the government typically will pay with securities to the purchasing bank the difference between the value of the assets and liabilities. The process usually involves the withdrawal or cancellation of the licence of the troubled bank, the termination of the owners' rights in the bank, the assumption of the troubled bank's deposits and good assets, and the takeover of the bank's problem assets by the resolution authority. In the UK a similar procedure is known as <i>private sector purchaser (PSP)</i> .
Bridge bank	A form of P&A, it involves the use of a temporary financial institution to receive and manage the good assets of one or several failed institutions. A bridge bank may be allowed to undertake some banking business, such as providing new credit and restructuring existing credits. A bridge bank does not require the authorities or state to acquire the shares of a failed bank, merely part or all of its property.
Deposit payoff	When the deposit insurer makes sure that customers of a failed bank receive the full amount of their insured deposits.
Open bank assistance	Open bank assistance (OBA) occurs when a distressed financial institution remains open with government financial assistance. This operation is similar to <i>temporary public ownership (TPO)</i> in the UK (although the TPO tool involves the UK Government acquiring all the shares of the failing bank).
Good bank/bad bank	When a troubled bank is split into a 'good bank', which continues to operate and it is typically acquired by a healthy bank, and a 'bad bank', which goes into administration. Insured deposits usually go with the 'good bank'. All other creditors typically travel with the 'bad bank' and become claimants in the insolvency procedure. This is also known as <i>partial property transfers (PPT)</i> .
Nationalisation	Nationalisation occurs when the government assumes ownership of an institution.
Bail-out	When a troubled bank receives financial help or liquidity from outside investors or from public funds in order to avoid bankruptcy.
Bail-in	A resolution mechanism that gives regulators the ability to impose losses on bondholders while ensuring the critical parts of the bank can keep running.

Source: Hoelscher and Quintyn (2003); Brierley (2009); McGuire (2012).

The toolkits designed to assist authorities in resolving troubled banks are generally country-specific, as are the various steps in the resolution of a troubled bank. Within each country, the legal framework defines the organisations that must be involved in the restructuring operations, their roles and their powers.

It is a generally accepted principle that the resolution should be carried out in a manner that minimises the cost for the resolution authority or the government (**least-cost resolution**). To determine the least-cost resolution, authorities will have to compare the different resolution mechanisms, taking into account the macroeconomic and market conditions at the time, which may render some of the options unfeasible. The Financial Stability Board (2011a) has issued 'Key attributes of effective resolution regimes for financial institutions' to promote effective and consistent implementation across countries and to support regulatory authorities to resolve financial institutions 'in an orderly manner without taxpayer exposure to loss from solvency support, while maintaining continuity of their vital economic functions'. The implementation of the 'Key attributes' is particularly relevant for cross-border bank resolution and the resolution of SIFIs (see Section 8.4.2). Box 8.4 outlines the aims of an effective bank resolution regime.

BOX 8.4 AIMS OF AN EFFECTIVE BANK RESOLUTION REGIME

An effective resolution regime (interacting with applicable schemes and arrangements for the protection of depositors, insurance policy holders and retail investors) should:

- (i) ensure continuity of systemically important financial services, and payment, clearing and settlement functions;
- (ii) protect, where applicable and in co-ordination with the relevant insurance schemes and arrangements such depositors, insurance policy holders and investors as are covered by such schemes and arrangements, and ensure the rapid return of segregated client assets;
- (iii) allocate losses to firm owners (shareholders) and unsecured and uninsured creditors in a manner that respects the hierarchy of claims;
- (iv) not rely on public solvency support and not create an expectation that such support will be available;
- (v) avoid unnecessary destruction of value, and therefore seek to minimise the overall costs of resolution in home and host jurisdictions and, where consistent with the other objectives, losses for creditors;
- (vi) provide for speed and transparency and as much predictability as possible through legal and procedural clarity and advanced planning for orderly resolution;
- (vii) provide a mandate in law for co-operation, information exchange and co-ordination domestically and with relevant foreign resolution authorities before and during a resolution;
- (viii) ensure that non-viable firms can exit the market in an orderly way; and
- (ix) be credible, and thereby enhance market discipline and provide incentives for market-based solutions.

Source: Financial Stability Board (2011a).

8.4.1 Bank resolution techniques

A resolution procedure is initiated when a bank is no longer viable (or likely to be no longer viable). A number of techniques can be used for dealing with problem banks; these tools may in some cases be alternatives and in other cases complementary. These are:

- liquidation (bank insolvency procedure);
- mergers and acquisitions (M&As);
- purchase and assumption (P&A);
- bridge bank;
- open bank assistance (OBA).

8.4.1.1 Liquidation (bank insolvency procedure)

When a bank is no longer viable, if a rescue is not possible or there is no prospect of restoring its longer-term viability, the bank may have to be liquidated under the country's general insolvency framework or a bank **special resolution regime (SRR)**. Depositors are likely to be paid off and creditors will be paid according to the hierarchy of claims. The resolution authority will dispose of all of the failed bank's assets, which can be a costly and time-consuming process. In addition, assets sold in a fire sale will yield less than their fair market value. A liquidation or bank bankruptcy procedure is most likely to be used for a bank whose failure is unlikely to have adverse systemic consequences and for which there is unlikely to be a buyer.

8.4.1.2 Mergers and acquisitions (M&As)

In the context of bank resolution, a merger of a troubled bank occurs when all the assets and liabilities of the firm are transferred and absorbed into another (healthy) institution. Mergers can be unassisted (voluntary) or government assisted. In an *unassisted merger*, regulators encourage the deal without providing any financial assistance. The plus side of this strategy is that there is no cost for regulators. A drawback is that the acquisition of a weak bank results in the weakening of an initially strong acquirer bank and therefore may leave regulators with a much larger weak bank to resolve.

In *government assisted mergers*, the government will offer some form of direct financial assistance in order to find a buyer for the troubled bank. The provision of direct financial assistance needs to be carefully structured so as not to benefit stockholders of the acquired institution at the expenses of taxpayers.

8.4.1.3 Purchase and assumption (P&A)

Purchase and assumptions are one of the most efficient methods for resolving troubled banks. A P&A involves healthy financial institution 'purchasing' some or all of a failed institution's assets and 'assuming' some or all of the institution's liabilities, usually insured deposits (where there is explicit deposit insurance) and potentially all deposits and even other liabilities (McGuire, 2012).

There are many types of P&A transactions. In a basic P&A transaction, the assuming institution generally takes on only limited assets, usually cash and cash equivalents. The liabilities are then matched to the assets taken and consist of either all or some of the deposits. In a 'whole bank P&A' the acquirer purchases the entire portfolio of the failed bank on an 'as-is' basis with no guarantees. P&A operations can also include some form of put option, entitling

the acquiring bank to return certain assets within a specified time period, or a contractual profit- or loss-sharing agreement related to some or all of the assets.

8.4.1.4 Bridge Bank

A **bridge bank** is a variation of P&A. In this scenario, the resolution authority acts as the acquirer by creating a new, temporary, full-service bank that is designed to bridge the gap between the failure of a bank and the time when the resolution authority can implement a satisfactory acquisition by a third party (McGuire, 2012). This resolution is used when the failure of a bank is unexpected (for example, in the case of the discovery of a fraud or the onset of a liquidity crisis). The bridge bank is designed to provide an immediate change in ownership and it is usually operational for a limited period of time (normally one or two years).

Transfer to a bridge bank may be the best option if a private sector solution requires more time to arrange, for example because potential **private sector purchasers** (PSPs) need to carry out further due diligence on the bank's books.

8.4.1.5 Open bank assistance (OBA)

Under certain circumstances, the government or the resolution authority may provide direct financial assistance to a bank in danger of failing. This 'assistance' may take the form of a loan, an asset purchase, a capital injection or the purchase of other debt instruments. In some circumstances the regulatory authorities might opt to exercise regulatory forbearance, particularly with respect to capital adequacy. In some specific circumstances, it may be necessary to take a bank into **temporary public ownership (TPO)** if there is no reasonable prospect of selling it to a PSP (either directly or through a bridge bank) and in the short run the bank's failure could represent a serious threat to financial stability. Table 8.3 summarises the pros and cons of the different resolution tools. Regulators will have to make decisions based on the bank financial condition, the reasons behind the financial troubles, market conditions at the time and the risk to systemic stability posed by the event of bank failure.

8.4.2 Resolution of large and complex financial institutions

The resolution methods discussed in Section 8.4.1 are not designed to cope with the failure of large and complex financial institutions (LCFIs). LCFIs (also called systemically important financial institutions – SIFIs) are large financial institutions that operate across borders and in many business areas; they have complex capital structures and can be funded by multiple types of liabilities (the definition of SIFIs was discussed in Chapter 7).

The first issue concerning the resolution of a LCFI is that it is likely to stretch government skills and resources. The second crucial issue is how to handle the resolution of large and internationally active banks. LCFIs are usually cross-border banking groups, whose resolution can be hampered by the fact that countries have their own SRRs, and SRR tools cannot automatically be applied extraterritorially.

A third problem is that existing SRRs are not designed to cope with the failure of investment banks (for example, a standard SSR could not effectively and orderly run down massive, complex derivatives or trading portfolios). As demonstrated by the Lehman Brothers and AIG episodes, unwinding a derivatives book is particularly complicated, as it requires dynamic hedging to preserve value, and that is very difficult to achieve in liquidation proceedings when counterparties take flight and funding dries up.

Table 8.3 Bank resolution methods

Resolution methods	Benefits	Costs
Liquidation (bank insolvency procedure – BIP)	Customers with insured deposits receive money quickly from the deposit insurance fund	- Customers with uninsured deposits and creditors have to wait for the (often lengthy) proceeds of the liquidation - Customers with uninsured deposits may not be paid the full uninsured amount - Customers must find a new bank - It is usually considered a ‘last resort’ because of high costs involved
Mergers and acquisitions (M&As)	- Unassisted mergers come at no cost to the authorities - There is no interruption to banking services	Healthy banks can become overburdened with the problems of the troubled bank
Purchase and assumption (P&A)	- Customers with insured deposits suffer no losses - Acquiring bank has the opportunity for new customers	- The majority of the assets might need to be liquidated - Uninsured depositors may suffer losses
Bridge bank	- Gives regulators time to arrange a permanent transaction - Gives purchaser time to assess the bank’s condition	- Duplicates part of the resolution process - Regulator becomes responsible for the operation of the bridge bank, which can be labour intensive and time consuming - May require ongoing liquidity support from the government - Difficult to retain best employees and customers - If a subsequent sale is not made, operating a bridge bank may exceed the costs of liquidation
Open bank assistance (OBA)	- Can be implemented relatively quickly - Could prevent systemic issues - Assets are kept in the private sector	- Promote a belief in the TBTF - Government funds could benefit private shareholders

Major reforms are currently being considered domestically and internationally to address these problems, in order to safeguard the stability of the financial system in the case of a LCFI failure. Key regulatory reforms are designed to reduce the systemic risk contribution of LCFIs and include proposals to impose additional capital charges on SIFIs (see Section 7.11.4); as well as proposals to facilitate the resolution of cross-border institutions. Regulatory measures that affect the structure, organisation or scope of the activities of LCFIs are also being discussed. Table 8.4 summarises these key regulatory initiatives at the national and international level, and the proposed timeline to implementation.

The measures illustrated in Table 8.4 aim to reduce the systemic risk contribution of SIFIs and to introduce measures to improve regulators’ capacity to resolve SIFIs.

Table 8.4 Regulatory initiatives

Key reforms	Aim	Timeline
Global reforms		
Basel IV (finalising Basel III)	• A standardised floor	2023
	• A reduction in standardised risk weights	2023
	• A higher leverage ratio G-SIBs	2023
Basel III capital standards	• Changes the definition of capital	2019
Basel III capital charges	• Better valuation of risk	2019
	• Incremental risk charge for trading book activity	
	• Higher capital charges for counterparty exposures in derivatives and repo trading	
	• Additional capital surcharge for G-SIFIs	
	• Capital charge assessed on (clearing member) banks' central counterparty default fund exposures	
G-SIBs surcharge	• Additional amount of common equity for systemically important banks	2019
Basel III liquidity requirements	• Liquidity coverage ratio: requires high-quality liquid assets sufficient to meet 10 days' outflows	2015
	• Net stable funding ratio: requires better maturity matching of assets and liabilities	2018
Basel III leverage ratio	• Sets a ceiling on the measure of exposures (regardless of risk weighting) against capital (3 per cent of Tier 1 capital over total exposure)	2019
FBS compensation guidelines	• Responsibility of boards for compensation policies	Implemented
	• Compensation should be aligned with risks and time horizons	
	• Supervisors should monitor compensation policies	
Corporate governance	• Emphasis on robust corporate governance, including the role of banks' board	
Resolution of G-SIFIs	• Reduce the likelihood that G-SIFIs will need to use public funds when they fail	
National reforms		
Volker Rule (Dodd–Frank Act) (US)	• Deposit-taking institutions restricted from trading activities, ownership of private equity and hedge funds	Implemented
Vickers Report (UK)	• Ring-fencing of UK retail banks from investment banking activities	2019
	• Additional capital for ring-fenced entity	
Bank Recovery and Resolution Directive (2014/59/EU)	• Establishing a common framework for the recovery and resolution of credit institutions and investment firms	Implemented
	• Institutions required to draw up recovery plans	
	• Resolution authorities to prepare resolution plans for each institution	
	• Enables resolution authorities to use the bail-in tool	
	• Member states, to set up <i>ex-ante</i> resolution funds	

Source: Adapted from IMF (2012c) and authors' updates.

8.5 Banking crises

When the default or failure of a bank brings about a loss of confidence in the banking system that leads to a run on banks, as individuals and companies withdraw their deposits, authorities are faced with a banking crisis. In the past 20 years, several countries suffered systemic banking crises of different severity. Leaven and Valencia (2012) identify 147 banking crises over the period 1970–2011. They also count 218 currency crises and 66 sovereign crises over the same time period.

During the period 1980–1993, liberalisation and deregulation led to the failure of 1,300 US ‘thrifts’ and 1,500 commercial banks (the so-called savings and loan crisis). During the mid-1990s, the Swedish, Norwegian and Finnish banking systems went from deregulated boom to bust, rescued by the state by means of varying forms of government indemnity, and final recovery. There were no bank defaults but the cost to taxpayers was high. Altogether the banking crises and the consequent support measures had a profound impact on the three countries’ economies, more particularly for Finland and Norway than for Sweden.

The French crisis of the early 1990s was centred mainly on real estate, causing the failure of seven rated banks. From the mid-1990s onwards the Japanese banking system also experienced a period of crisis, with no bank defaults. The enormous inflation and subsequent crash (in the early 1990s) in Japanese real estate prices severely damaged the major banks. At no period in their post-Second World War history have the Japanese banks needed government support more than from the mid-1990s to the present day (see also Chapter 16 for more details on the Japanese banking crisis).

Bulgaria, the Czech Republic, Hungary, Poland, Romania and Slovakia experienced banking crises during the 1990s, particularly during the period of transition from communism to capitalism and prior to the EU accession.

In 1997, a number of emerging economies in South East Asia were affected by a severe financial crisis, which started in Thailand and then spread to Indonesia, Malaysia, South Korea and Vietnam.

In more recent times, the US and UK experienced a banking crisis in 2007, which then spread to a number of other countries between 2008 and 2009.

The literature indicates that banking crises happen in waves, often preceded by a credit boom. Figure 8.4 illustrates the cycles of banking crises over the past 40 years.

Governments, central banks and external agencies have dealt with banking crises in a number of ways, often according to the circumstances, as described in Section 8.4. Although there is no unique recipe, there are some common ingredients to successful crisis management:

- governments must be willing to recognise the scale of the problem as soon as possible;
- governments should support supervisory authorities who want to close insolvent banks;
- governments should be willing to commit substantial fiscal resources to the banking system;
- transparent actions with regards to NPLs should be adopted at an early stage;
- improved regulatory and supervisory frameworks are often necessary.

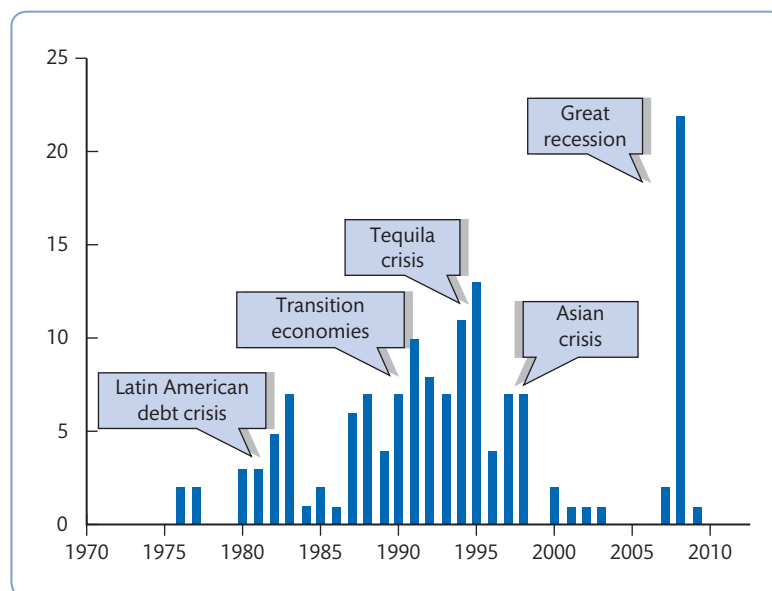


Figure 8.4 Banking crises

Source: Laeven and Valencia (2012), p. 10.

8.5.1 Systemic banking crises

The success of crisis management and bank restructuring depends, ultimately, on a favourable macroeconomic environment and the wherewithal of the authorities to make hard (often politically unpopular) decisions regarding banking system restructuring. Sometimes though, things do not go according to the regulators' plans and the failure of one or more banks triggers a number of defaults with profound consequences on the economy. The failure of a key bank can prompt runs on other banks as bank customers withdraw their savings, unable to distinguish between sound and troubled institutions. Further, because of the interconnectedness of banks, the failure of one institution can immediately affect others. This is known as *bank contagion* and may lead to *bank runs*.

Banking systems are vulnerable to *system risk*, which is the risk that problems in one bank will spread through the whole sector, and crises can become 'systemic'. This vulnerability to systemic risk is one of the main reasons for bank regulation (see Chapter 7).

Systemic banking crises are rare events, according to Boissay *et al.* (2013), who estimate that they occur on average every 40 years. However, they cause recessions that are deeper and last longer than other recessions.

According to Laeven and Valencia (2012), banking crises can be defined as 'systemic' if there are: (1) significant signs of financial distress in the banking system (as indicated by significant bank runs, losses in the banking system and/or bank liquidations); or (2) significant banking policy intervention measures in response to major losses in the banking system. More specifically, Laeven and Valencia (2012) consider a crisis episode to be systemic if at least three out of the following six measures have been used by regulators: (i) extensive liquidity support (5 per cent of deposits and liabilities to non-residents); (ii) bank restructuring costs (at least 3 per cent of GDP); (iii) significant bank nationalisations; (iv) significant guarantees put in place; (v) significant asset purchases (at least 5 per cent of GDP); and (vi)

deposit freezes and/or bank holidays. Using these metrics, they identify 17 countries that experienced a systemic banking crisis between 2007 and 2011; with eight more countries experiencing a borderline case (i.e. less than three of the above measures were used). In the past, however, some countries intervened in their financial sectors using a combination of less than three of these measures listed above, but on a large scale (for example, by nationalising all major banks in the country). Therefore, a banking crisis can be considered systemic if: (i) a country's banking system exhibits significant losses resulting in a share of NPLs above 20 per cent; (ii) bank closures of at least 20 per cent of banking system assets; or (iii) fiscal restructuring costs of the banking sector are sufficiently high, exceeding 5 per cent of GDP.

Over the past 20 years, a number of economies have suffered systemic banking crises, with far-reaching negative economic implications. These crises resulted in substantial losses in terms of wealth, output and jobs. For example, the average cumulative output losses in Argentina (2001–2002), Indonesia (1998) and Turkey (1999–2001) have been estimated in the range of 12–15 per cent of GDP. The economic cost of the new crises is on average much larger than that of past crises, both in terms of output losses and increases in public debt, as illustrated in Table 8.5. These differences in part reflect an increase in the size of financial systems and the fact that the 2007–2009 crisis was concentrated in high-income countries, whereas previously banking crises seemed to affect mainly developing and transition economies (with some notable exceptions, such as the Japanese banking crisis).

Table 8.6 compares the costs, in terms of output losses and fiscal costs, of the two largest and most recent crises, the 1997 Asian crisis and the 2007–2009 crisis. In 1997, a number of South East Asian economies suffered a severe financial crisis causing substantial output losses. The 2007–2009 financial crisis, or global financial crisis, triggered the deepest recession since the Second World War and has become known as '*The Great Recession*'. The consequences of the Covid-19 pandemic on the global economy are still unknown – as countries are focusing resources on vaccination programmes, the economic cost will only become clear once the pandemic is over.

8.5.2 Financial crises

These recent crises episodes have not been confined to the banking sector, but have involved other financial markets. As a consequence, some economists broaden the definition of financial crisis to include features of financial fragility, bank panics and contagion, when financial

Table 8.5 The costs of banking crises

	Direct fiscal costs	Increase in public debt	Output losses
Previous crises (1970–2006)	Medians (% of GDP)		
Advanced economies	3.7	36.2	32.9
Emerging markets	11.5	12.7	29.4
All	10.0	16.3	19.5
Recent crises (2007–2009)	Medians (% of GDP)		
Advanced economies	5.9	25.1	24.9
Emerging markets	4.8	23.9	4.7
All	4.9	23.9	24.5

Source: Adapted from Laeven and Valencia (2014).

Table 8.6 Comparing the costs of the Asian and global crises

	Country	Year start	Year end	Output loss	Fiscal costs (% of GDP)	Fiscal cost (% of financial sector assets)	Peak NPLs
Asian crisis	Indonesia	1997	2001	69.0	56.8	105.4	32.5
	Malaysia	1997	1999	31.4	16.4	12.7	30.0
	Thailand	1997	2000	109.3	43.8	30.6	33.0
	Overall			69.9	39.0	49.6	31.8
Global financial crisis	United Kingdom	2007	2011	25.3	8.8	5.8	4.0
	United States	2007	2011	31.0	4.5	2.2	5.0
	Benelux	2008	2012	26.0	8.8	1.7	2.5
	Germany	2008	2009	12.3	2.7	0.9	3.7
	Greece	2008	2012	64.9	28.7	17.1	37.1
	Iceland	2008	2012	34.5	37.6	14.3	61.2
	Ireland	2008	2012	107.7	37.6	4.5	25.7
	Spain	2008	2012	38.8	5.4	2.0	9.4
	Overall			42.6	16.8	6.1	18.6

Notes: Output loss in % of GDP. Output losses are computed as the cumulative sum of the differences between actual and trend real GDP over the period $[T, T + 3]$, expressed as a percentage of trend real GDP, with T the starting year of the crisis. Fiscal costs are defined as the component of gross fiscal outlays related to the restructuring of the financial sector. They include fiscal costs associated with bank recapitalisations but exclude asset purchases and direct liquidity assistance from the treasury. NPL as expressed in % of total loans. NPLs data come from IMF Staff reports and Financial Soundness Indicators. The duration of crises is truncated at 5 years, starting with the first crisis year.

Source: Authors' calculations. Database made available by Laeven and Valencia (2020) 'Systemic banking crises database II'.

markets experience volatility and financial firms suffer illiquidity and insolvency. In addition to banking crises, financial crises can come in many forms. Reinhart and Rogoff (2009) distinguish two types of crises: (i) *currency* and *sudden stop* crises and (ii) *debt* and *banking* crises.

A **currency crisis** involves a speculative attack on the currency, resulting in a devaluation or sharp depreciation. This will force the authorities to defend the currency by selling foreign exchange reserves, raising domestic interest rates or imposing capital controls.

Currency crises have been relatively common in the past; the most remarkable episodes include the crisis of the British pound in 1976 and the European Exchange Rate Mechanism (ERM) crisis in 1992–1993. A sharp depreciation from a fixed exchange rate was also central to the Mexican peso crisis (which became known as the Latin American Tequila Crisis) in 1994–1995 and the Thai crisis of 1997, which marked the beginning of the Asian financial crisis of 1997–1998.

A **sudden stop** (also known as a capital account or balance of payments crisis) can be defined as a sudden (and often large) decrease in international capital inflows or a sharp reversal in aggregate capital flows to a country, likely taking place in conjunction with a sharp rise in its credit spreads. Sudden stops are usually followed by a sharp decrease in output, private spending and credit to the private sector, and real exchange rate appreciation. Sudden stops hit many emerging economies, particularly in the aftermath of financial crisis.

Claessens and Kose (2013) define a *foreign debt crisis* as an episode of financial turbulence that takes place when a country cannot (or does not want to) service its foreign debt. It can take the form of a *sovereign debt crisis* or *private debt crisis* (or both in some cases). The most recent episode is the European sovereign debt crisis, which started in 2008 with the collapse of Icelandic banks (see Box 8.5), and spread to Greece, Ireland and Portugal and

BOX 8.5 THE ICELANDIC MELTDOWN

Relative to the size of its economy – Iceland’s population is just over 300,000 – the collapse of Iceland’s banking system in autumn 2008 has been adjudged the largest of all time by the IMF. Following financial deregulation in 2001, Iceland’s three major banks, Landsbanki, Kaupthing and Glitnir, developed a business model that circumvented the constraints on growth implied by the small size of the Icelandic economy, by attracting funding from international capital markets. Between 2006 and 2008 Landsbanki and Kaupthing set up online banking operations offering high-interest internet accounts to depositors in the UK and the Netherlands in the case of Landsbanki’s Icesave brand, and through subsidiaries trading under the Kaupthing Edge brand in nine European countries.

Prior to the banking crisis, Iceland’s current account deficit was clearly unsustainable, having reached 25% of GDP in 2006, and 15% in 2007. Between January and September 2008, consumer price inflation was running at around 14%, and domestic interest rates reached 15.5%. Despite a 35% decline in value against the euro during the first nine months of 2008, the krona was still significantly overvalued, bolstered by short-term capital inflows attracted by the high domestic interest rate. When liquidity in the interbank markets dried up in mid-September 2008 following the liquidation of Lehman Brothers in the US, the Central Bank of Iceland had inadequate reserves to be able to guarantee the banks’ debts as lender of last resort. The European Central Bank, US Federal Reserve, Bank of England and the three Nordic central banks collectively declined to provide sufficient assistance to avert the imminent crisis.

On 29 September 2008 it was announced that the Icelandic government was to acquire a 75% stake in Glitnir. This part-nationalization was not completed, however, and a few days later Glitnir was placed into receivership. Reports in the British press over the weekend of 4–5 October 2008 appear to have triggered a run on savings in Icesave by UK and Dutch online depositors, and Landsbanki was placed into receivership on 7 October. Since Icesave was a branch of Landsbanki, its UK depositors were not protected under UK deposit insurance; however, on 8 October the UK government froze Landsbanki’s UK assets, and announced it would compensate UK retail depositors in full. A number of UK local authorities

and other governmental organisations, which had deposited spare funds with Icelandic banks, would not be guaranteed reimbursement. Meanwhile, on 8 October the UK’s Financial Services Authority placed Kaupthing’s UK subsidiary into administration and sold its internet bank Kaupthing Edge to the Dutch group ING Direct. In Iceland Kaupthing followed into receivership on 9 October, and over the next few days Kaupthing’s other subsidiaries were either wound up or taken into public ownership by the respective national authorities.

With the Krona continuing to fall precipitously against the euro, the Icelandic government applied for IMF assistance in late-October 2008. In November the IMF agreed to provide a \$2.1bn standby programme over two years, supplemented by assistance in the form of loans and currency swaps from the governments of the Nordic countries, Russia, Poland, the UK, the Netherlands and Germany, which brought the headline value of the full package to over \$10bn. The terms of the IMF package impose obligations on the Icelandic government in the areas of currency stabilisation and inflationary control, bank restructuring and fiscal retrenchment. A sharp fall in GDP was anticipated in 2009, projected at around 10% at the time of writing. Several years of austerity appeared inevitable.

A new Icelandic government elected in April 2009 was committed to applying for full EU membership and adoption of the euro as soon as possible. Membership of a global reserve currency was viewed as offering future protection against the exposure that destroyed the Icelandic banking system in 2008, and may alleviate the banks’ difficulties in raising short-term foreign funds to cover their foreign debts. Iceland applied to join the EU on 16 July 2009. However, contentious issues, particularly relating to fisheries, remained unresolved. On 13 September 2013 the Icelandic government dissolved its accession team and suspended its application to join the EU.

By 2014, around £1 billion of cash tied up in the failed Icelandic banks had been recovered by UK local authorities. Following the sale of the remaining claims against Landsbanki (LBI), Glitnir, Heritable and Kaupthing Singer & Friedlander, UK councils hoped to recover 90p for every pound invested.

Source: Adapted from Goddard *et al.* (2009). Authors’ own updates.

subsequently to Italy and Spain, leading to a crisis for the eurozone (the eurozone troubles will be discussed in more detail in Chapter 14).

Crises episodes often occur together, currency crises followed by banking crises (for example in the case of the Asian crises), or banking crises followed by sovereign debt crises like in the case of the eurozone crisis. These occurrences are defined as **twin crises**. Rarer are episodes of '*triplet crises*' (i.e. the simultaneous occurrence of three or more crises). Since the early 1970 to date, Leaven and Valencia (2020) have identified 151 episodes of banking crises, 91 debts crises and 236 episodes of currency crises. They also identified 54 episodes of twin crises, of which 31 were the result of the simultaneous occurrence of currency and banking crises, 20 currency and debt crises, and only 3 banking and debt crises. Finally, they identified only 11 episodes of triplet crises since 1970.

8.5.3 Identifying the causes of banking and financial crises

The identification of the causes of banking and financial crises is important in terms of crisis management and effectiveness of the proposed solutions. The nature of the underlying causes may have important bearings in the optimal official response. There is a vast literature on the causes of banking crises; this literature is fairly divided between advocates of microeconomic factors and supporters of macroeconomic reasons.

The often-cited microeconomic reasons include:

- poor banking practices (inadequate capital, inadequate credit risk assessment resulting in non-performing loans, insufficient diversification of the lending portfolio, excessive mismatching of maturity and currency);
- principal–agent incentive problems (particularly when loan officers are rewarded on the volumes of loans granted);
- over-staffing (particularly in state-owned banks);
- restrictive labour practices (sometimes delaying the adoption of IT).

Macroeconomic reasons, although not relieving bank management of their responsibilities, are often seen as a catalyst of crises. Macroeconomic shocks, such as the oil crisis in the 1970s, can strain even properly managed banks.

A third set of causes is the so-called system-related, in the sense that the environment is not conducive to the development of an efficient banking sector. For example:

- large state ownership in the banking sector can distort the industry. If state-owned banks enjoy special privileges, this may distort competition and limit banks' diversification possibilities;
- government direction of credit may prevent banks from developing credit risk management skills;
- restrictions on foreign banks entry;
- poor market discipline (due to moral hazard and excessive deposit insurance);
- weak corporate governance;
- poor supervision;
- an inadequate legal framework may limit the effectiveness of the banking system;

- an underdeveloped securities market may concentrate too much risk on the banking system;
- high market power of incumbent banks, leading to lack of competition.

Banking crises may result from rapid changes in the environment in which banks operate. For example, in the early 1990s Mexico experienced a rapid privatisation process of its commercial banks, coupled with financial liberalisation measures and sudden reduction of the borrowing requirement of the public sector. The rapid expansion of credit that followed these changes, coupled with weak supervision, led to a financial sector crisis in 1994.

Much has been written on the causes of the 2007–2009 global financial crises. While academics and policy makers' views may differ on the importance of different factors, most list the following as the key reasons (Claessens and Kose, 2013):

- 1 asset price increases that turned out to be unsustainable;
- 2 credit booms that led to excessive debt burdens;
- 3 build-up of marginal loans and systemic risk; and
- 4 the failure of regulation and supervision to keep up with financial innovation.

These triggers have been present even in previous crises. However, the 2007–2009 global financial crises was also brought about by new factors, including: (1) the widespread use of complex and opaque financial instruments; (2) the increased interconnectedness of financial markets, nationally and internationally; (3) the high degree of leverage of financial institutions; (4) the central role of the household sector; and (5) the emergence of an unregulated 'shadow banking' sector. These factors, in combination with the ones common to previous crises, led to the worst financial crisis since the Great Recession, the consequences of which are still being felt in many countries.

8.5.4 Crises management

Crisis management can be summarised in three distinct phases:

- 1 the containment phase;
- 2 the resolution phase;
- 3 the structural reform phase.

Crisis management starts with the *containment phase*, when the crisis is unfolding and a speedy regulatory response is critical. The first step is the control of liquidity pressures on the banking sector, via emergency liquidity lines to troubled financial institutions. In this phase, the main aim is to stabilise the markets, which sometimes entails government blanket guarantees to financial institutions.

In the *resolution phase* a broad range of measures (such as those described in Section 8.4.1) are taken to restructure banks. The final phase is the *reform phase*, when fundamental changes to the regulation and supervision of the financial sectors are enacted.

The specific actions taken by domestic and international regulatory bodies in response to the global financial crises are discussed elsewhere in this textbook. In general terms, the policy responses during the 2007–2009 crises were similar to those used in the past. The first

actions of many national central banks and governments were to ease liquidity pressures, through liquidity support and guarantees on bank liabilities. During the bank resolution phase, a broad array of resolution techniques was used, including asset purchases, asset guarantees and equity injections. All these methods, to some extent, have been used in the past. Some of the measures put in place have been more innovative (see, for example, the discussion of unconventional measures of monetary policy in Section 6.4.5). The current phase is the structural reforms phase, and the key reforms are summarised in Table 8.4.

One key feature of the 2007–2009 global financial crisis was that it predominantly affected advanced economies with large, internationally integrated financial institutions that were deemed too big and/or interconnected to fail. The large international networks and cross-border exposures of these financial institutions helped propagate the crisis to other countries. Failure of any of these large financial institutions could have resulted in the failure of other systemically important institutions, either directly by imposing large losses through counterparty exposures or indirectly by causing a panic that could have generated bank runs. This prompted unprecedented large-scale government interventions in the financial sector in many countries.

How successful have these interventions been? This is a difficult question to answer, as it is difficult to compare the success of crisis resolution policies given differences across countries and time in the size of the initial shock to the financial system, the size of the financial system, the quality of institutions, and the intensity and scope of policy interventions. As the Covid-19 pandemic continues to influence the economic outlook in many countries, the implications for financial stability are unclear. Prior to the pandemic, there was a consensus that banks were in a better position relative to 2008, due to the extensive regulatory changes following the global financial crisis. Many now fear a surge of non-performing loans, once moratoria and other support measures are unwound as economies start to re-open (Kasinger *et al.*, 2021).

8.6 Conclusion

Bank regulation cannot prevent financial crises, but the regulatory framework that is currently being shaped will influence the development of the banking system for many years to come. Bank resolution measures recently implemented by governments (including the purchase of impaired assets, the recapitalisation of troubled banks and liquidity injections into the financial system) have possibly avoided even more serious consequences, despite concerns that publicly funded bank bailouts sent the wrong signals to bank investors and executives who caused the problems. At the time of writing, the global economy is suffering due to the Covid-19 pandemic and the roadmap to recovery in Europe, the US and the rest of the world remains uncertain. There are fears that many EU banks still have to recover fully from the consequences of the 2007–2009 global financial and eurozone crises to withstand an increase in non-performing loans. Many predict that banks will become leaner, more capitalised, less leveraged and more heavily regulated than they have been in the past. If the banking system that emerges from the recent period of severe instability is more reliable and efficient, then the lessons of the 2007–2009 crises may have been learned.

Going forward, banks will have to face new challenges, due to the consequences of the Covid-19 pandemic and the risks posed by climate change.

Key terms

Bridge bank	Early warning system (EWS)	Peer group analysis	Stress testing
CAMELS	Least-cost resolution	Private sector purchaser	Sudden stops
Collateral	Non-performing loans (NPLs)	Special resolution regime (SRR)	Temporary public ownership (TPO)
Currency crisis			Twin crises

Key reading

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REVISION QUESTIONS AND PROBLEMS

- 8.1 The reasons why banks fail are numerous and often interlinked. Managerial deficiencies are often important and there is a very fine line between bad management and fraud. Discuss.
- 8.2 Discuss the different approaches to the resolution of a failed bank. Discuss the principle of 'least-cost resolution' and how it applies to the choice of regulatory tool.
- 8.3 Discuss the use of EWS in the context of the anticipation and prevention of banking problems. How useful are stress tests?
- 8.4 What are non-performing loans? Discuss the relevance of NPLs in the context of bank failure and bank restructuring.
- 8.5 Discuss the pros and cons of bank resolution tools, giving examples of successful and not so successful outcomes.
- 8.6 Discuss the different problems posed by the failure of a purely domestic bank and of an internationally active bank. What are the main issues associated with cross-border bank supervision?
- 8.7 Define the process of crisis management. What elements influence the success of crisis management and bank restructuring?
- 8.8 Briefly discuss major regulatory developments that have impacted LCFIs since 2007.